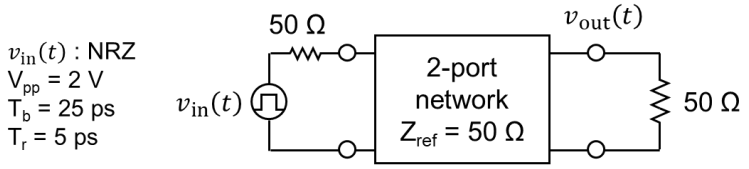
Homework: Peak distortion analysis (PDA)

Consider the following circuit. The s2p of the channel (2-port network) is provided to you. The input source $v_{\text{in}}(t)$ is a 40 Gbps NRZ signal with risetime 5 ps. Because of impedance matching, the transfer function from $V_{\text{in}}(f)$ to $V_{\text{out}}(f)$ is simply $S_{21}/2$.



1. Run PDA to obtain the worst-case eye boundaries at $v_{\text{out}}(t)$.
2. Run sectional convolution to obtain the single-colored eye and the statistical eye assuming $v_{\text{in}}(t)$ is purely random NRZ. Overlay the result with the worst-case boundaries obtained in step 1.
3. Run sectional convolution to obtain the single-colored eye and the statistical eye using the bit file '`HW6_wcbitpattern.txt`' for $v_{\text{in}}(t)$. Overlay the result with the worst-case boundaries.
4. Comment on the results.

Optional problems:

5. Run ADS to obtain the eye diagrams using random signal for $v_{\text{in}}(t)$ as well as the bit file.

Note:

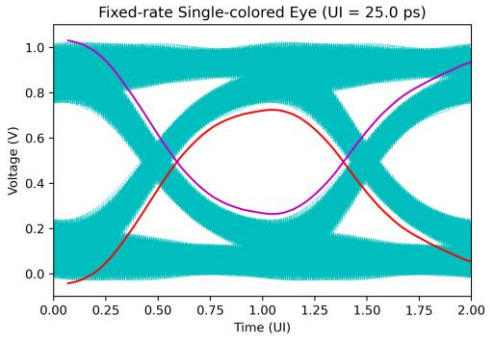
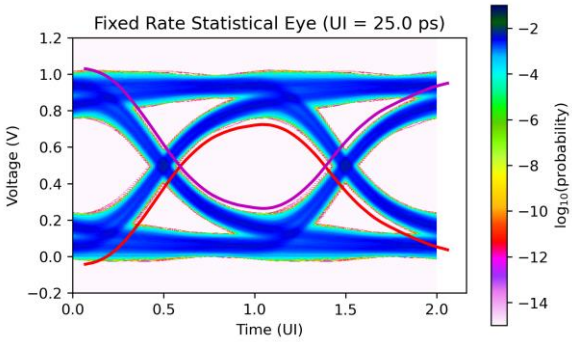
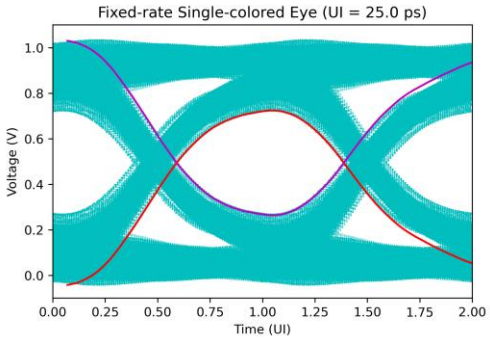
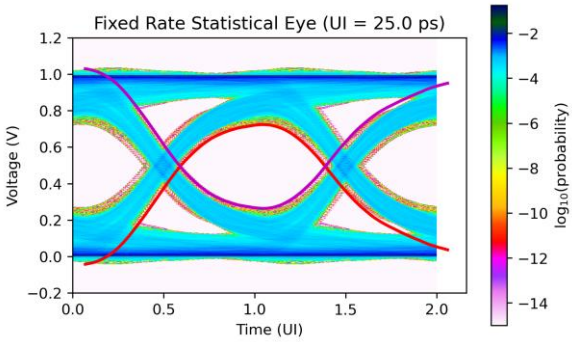
(1) To perform PDA, you first need to obtain the SBR. In this problem, SBR is obtained through IDFT. However, before feeding the SBR to PDA, it is recommended to truncate the SBR in advance. There are several reasons for doing this: efficiency, and avoid overestimating the worst-case.

(2) When implementing the PDA algorithm, we will set a tolerance (`tol`) level: if `abs(SBR[k]) < tol`, then we will not count `SBR[k]` when computing the worst-case boundaries. (Why do this?) Example of `tol` is 1 mV or 0.1 mV.

(3) In the PDA function, since we pick the main cursor at the point where SBR is maximum, the time axes of wc1 and wc0 may have a shift relative to the time axis of SBR. Therefore, when overlaying the worst-case boundaries on the eye diagram, remember to add a shift so that the results are comparable. The amount of shift is related to the variable 'kstart' returned by the PDA function.

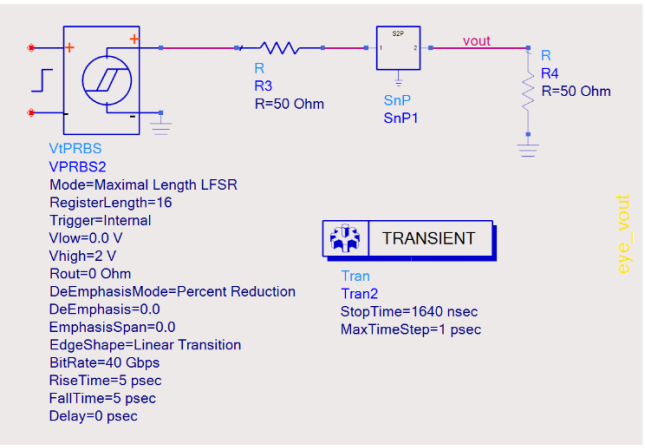
(4) In this homework, to properly compare with the PDA's results, you should plot the 1-UI eye, for both the single-colored one and the statistical one, and then duplicate the plot to make it 2-UI wide. If you directly construct the 2-UI eye, the correlation between the results will not be as good (why?).

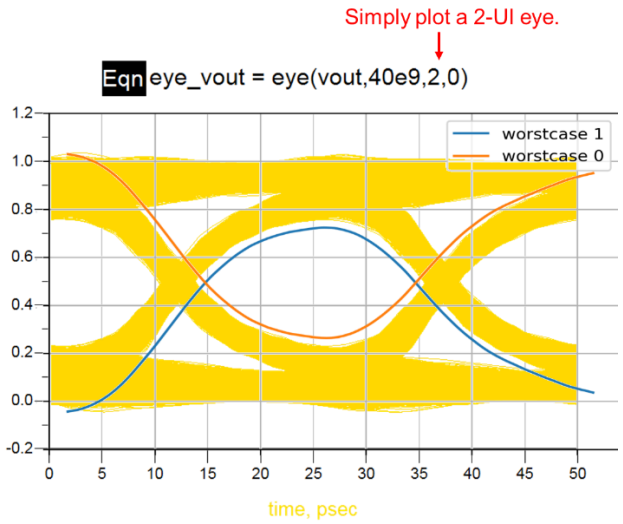
Appendix 1: Example answer

	Single-colored eye	Statistical eye
Purely random NRZ		
Bit pattern file		

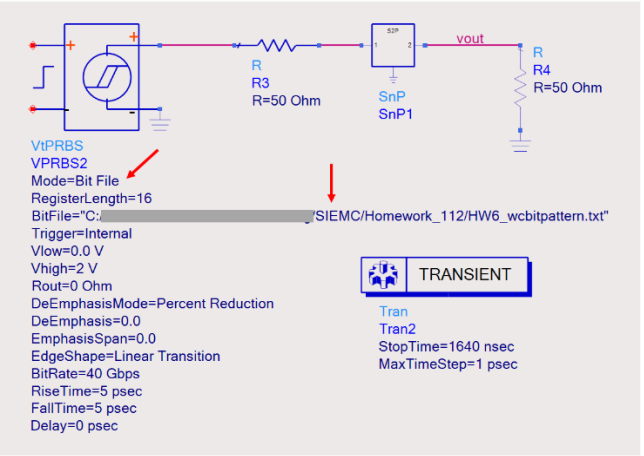
Appendix 2: ADS simulation setting

Random NRZ





Bit pattern file





Procedure to overlay PDA's result with eye diagram from ADS:

(1) Carefully set clear xlim and ylim in all plots.

(2) Copy the eye diagram and Python's PDA plot to powerpoint.

(3) Put the PDA's worst-case boundaries (Python plot) on top of the eye.

(4) Set the background color of PDA figure to transparent
(點選圖片➔圖片格式➔色彩➔設定透明色彩➔點選白底處)

(5) Resize the PDA figure such that it has the same scale (both vertically and horizontally) as the eye diagram figure.

(6) Horizontally shift the PDA figure to match the eye center (because the SBR simulation and eye diagram simulation may not have the same starting time).